

Supplementary Material

Appendix A. Modeling the stochastic processes

Figure A.1 displays sample realizations of the stochastic processes for the truck purchase prices by technology. The grey area in the charts shows the spectrum the stochastic processes cover within 100 iterations. The purchase price of ICEVs is exposed to opposing influences. On the one hand, most publications expect strict regulations to increase the price soon (Gnann et al., 2017; Kühnel et al., 2018; Giuliano et al., 2021). On the other hand, vital efficiency improvements may also induce a downward effect (Kasten et al., 2016; Mottschall et al., 2019; Hacker et al., 2014). Consequently, we model a relatively widespread development in the first half of the time frame with the tendency towards a cost increase. Apart from that, ICEVs will most likely drop in value when they are not produced or sold anymore (or even forbidden). The production of heavy-duty ICEVs is expected to cease around 2040 to 2045 (European Automobile Manufacturers' Association, 2020; Öko-Institut, 2021). A similar development is presumed for the purchase price of LNGVs. Typically, they are slightly more expensive than ICEVs because the expected smaller degree of market penetration leads to higher production costs per unit (Gnann et al., 2017; Kühnel et al., 2018; Mottschall et al., 2019; Kasten et al., 2016; Hacker et al., 2014). Naturally, BEVs start with a very high purchase price, but a steep decline is assumed until 2030. The reasons are, among others, various forms of funding, the achievement of greater technological maturity, and the associated higher market penetrations driving down production costs. After that, the cost decrease gets smaller, and eventually, a final value is reached. At this point, maturity and penetration reach their peak. We further assume that BEV purchase prices do not fall below current ICEV purchases prices because the technology will always remain a bit more expensive (mainly due to the cost-pushing battery) (Gnann et al., 2017; Giuliano et al., 2021; Kühnel et al., 2018;

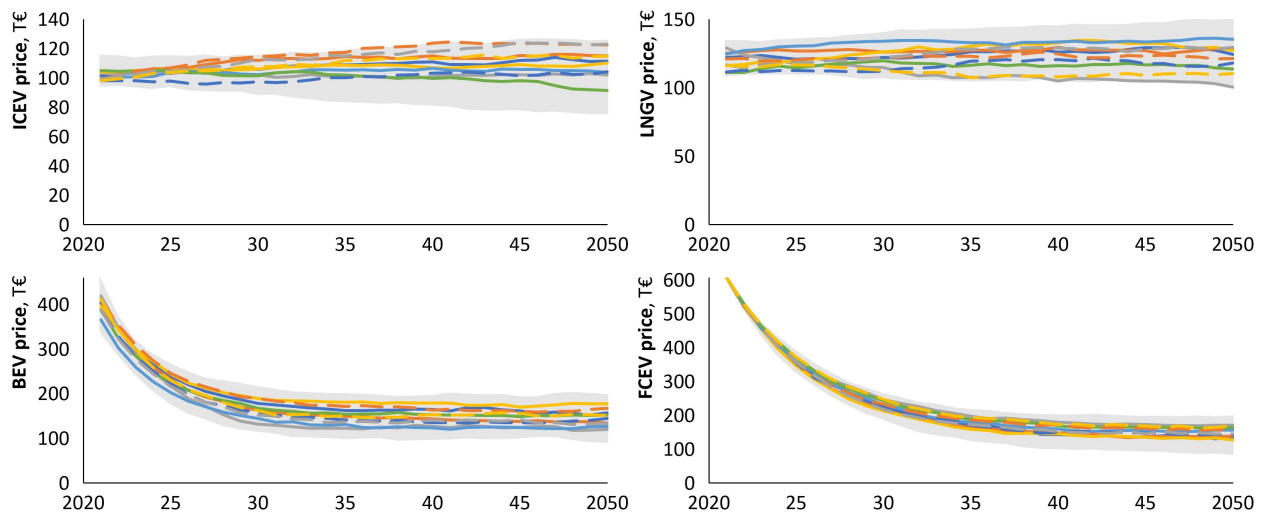


Figure A.1: Stochastic process spectrums (grey areas) and selected samples (lines) of the purchase prices for ICEVs (top left), LNGVs (top right), BEVs (bottom left), FCEVs (bottom right)

Unterlohner, 2020). We apply a somewhat positive interpretation (i.e., a faster decline and a lower final value) from the inspected literature as the current price developments hint in that direction. Nevertheless, we model the stochastic process with a big spread resulting from great uncertainty and low consensus in the literature. FCEVs start with an even higher purchase price than BEVs. The technological maturity and the associated market penetration are expected to develop much slower (Hacker et al., 2014; Kasten et al., 2016; Mottschall et al., 2019; Gnann et al., 2017; Kühnel et al., 2018; Unterlohner, 2020). One major concern is the missing infrastructure for the hydrogen distribution network and refill stations. Still, the mean reversion level is assumed to be reached around 2035 to 2040. At the moment, many projects that aim to build missing infrastructure to drive market readiness receive government approval (Weinhold, 2021). In the long run, the FCEV purchase price is assumed to sink below the BEV purchase price. The technologies are generally very similar, but the battery and related components will presumably be more expensive than the fuel cell and the hydrogen tank upon reaching full maturity (Hacker et al., 2014).

The prices paid for the fuel alternatives according to multiple sample realizations of the stochastic processes can be seen in Figure A.2. Considering the reviewed literature, we anticipate a continuously increasing diesel price as Germany gradually stops to build on fossil fuels and regulations and carbon pricing take effect. The literature predominantly proposes a linear development until 2050 (Mottschall et al., 2019; Kühnel et al., 2018). In contrast, given the recent price developments, we assume that the increase will follow an exponential curve with a steep ascent in the first periods of the time frame. It shall be noted that we exclude any disruptive external effect due to events like the war in Ukraine on the mean reversion level. Drastic changes to the diesel price are revisited in the sensitivity analysis in section 5.3 of the main document. LNG prices are also expected to increase continuously based on the same reasoning (Mottschall et al., 2019; Kühnel et al., 2018). As the LNG fuel has a slight emissions advantage

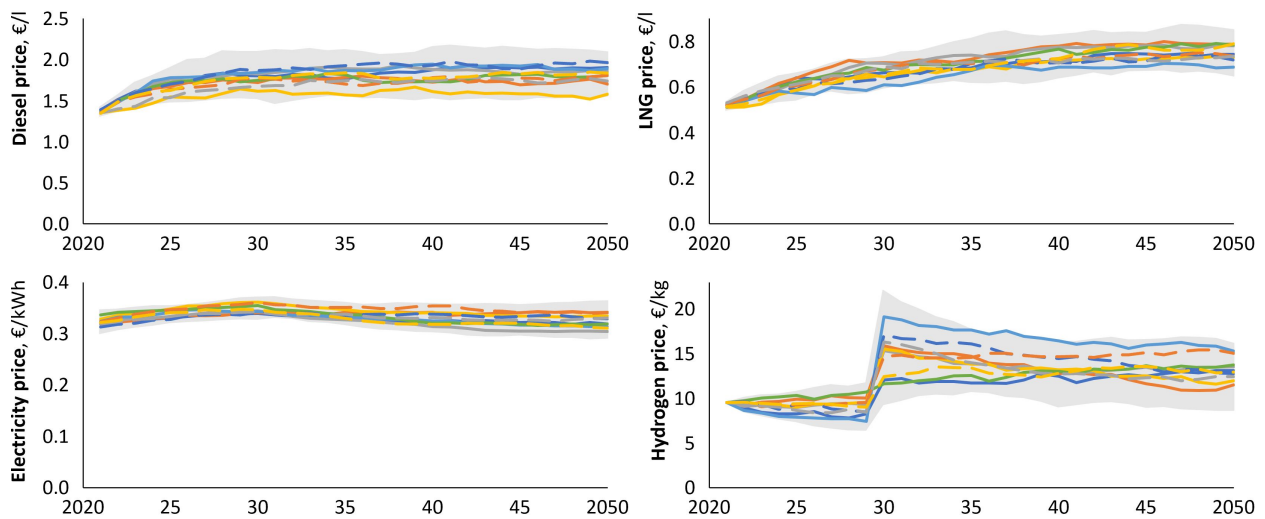


Figure A.2: Stochastic process spectrums and selected samples of prices for diesel (top left), LNG (top right), electricity (bottom left), hydrogen (bottom right)

and receives less government attention due to the lower market presence, we apply a much lower mean reversion speed. The price of electricity is assumed to be much more stable than the two aforementioned. A slight increase in prices is modeled until 2030 since the German government is expanding the renewable energy infrastructure (which is the case with or without the EEG reallocation charge). Afterward, the prices relax until 2050 and eventually reach a mean reversion level slightly below today's prices (Mottschall et al., 2019). Concerning the prices of hydrogen, it is necessary to consider the source or, rather, the production strategy behind it. We assume that until 2030 there will be no reliable offer of green hydrogen on the market. The prices start at the current market price and decline substantially due to increasing market penetration and a growing infrastructure (Mottschall et al., 2019; Kühnel et al., 2018). As of 2030, green hydrogen becomes increasingly available, and the case study company commits to using it. As a result, the hydrogen price jumps to a much higher level, from which it gradually but slowly decreases until 2050 (Prognos et al., 2020; Deutscher Bundestag, 2020).

Finally, Figure A.3 shows sample realizations of German nEHS and EU ETS carbon prices as well as the electric battery purchase price. The nEHS price is considered in multiple ways in the program. It impacts the prices of diesel, LNG, and hydrogen. Furthermore, it is used as a carbon price for the emissions of the case company in the target function. The German government predefined the development of the nEHS price until 2025. After 2025, the price is determined by the market, and the literature expects a very strong increase until 2050 (Umweltbundesamt & DEHSt, 2020; Kemmler et al., 2020; Hennes et al., 2021). Based on this, we assume the mean reversion level is reached around 2040. At this time, fossil fuels will likely have a stable low market share, resulting in the prices settling. The EU ETS carbon price directly influences the electricity price in the model. In recent years, the EU ETS price experienced a significant increase. Most of the reviewed literature did not foresee the extent of this development, so the

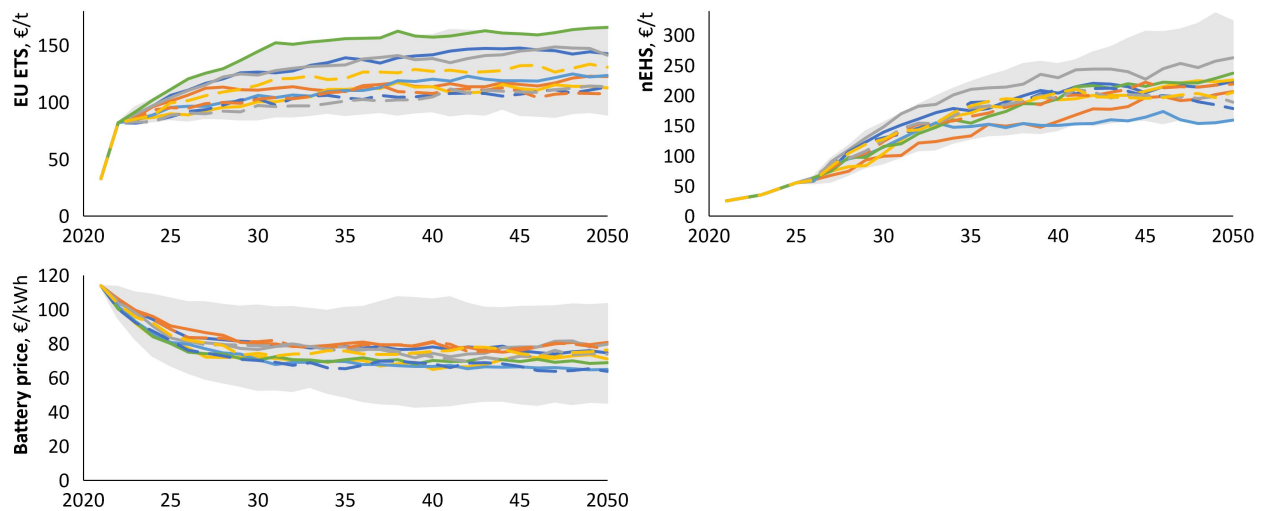


Figure A.3: Stochastic process spectrums and selected samples of EU ETS (top left), German nEHS (top right), electric battery purchase price (bottom left)

proposed values fall short (Kemmler et al., 2020; Pyrka et al., 2020; Prognos et al., 2021, 2020). We apply an extrapolation of the current real-world prices with the growth assumptions from the literature. The final stochastic process randomly develops in the corridor between this extrapolation and the original values given by the publications. Lastly, the electric battery purchase price becomes relevant when a battery in a BEV needs to be replaced. We assume this to occur after driving 500,000 to 600,000 kilometers (Kühnel et al., 2018). The fuel cell of an FCEV is expected to last for the whole life of the FCEV. Hence, it does not need to be replaced beforehand (Gnann et al., 2017). Mauler et al. (2021) discuss that most subject-matter publications do not accurately forecast the purchase prices of electric batteries. The prices declined much faster than most of the literature presumed. According to them, He et al. (2020) produced the best estimate, so the modeled stochastic process is predominantly fitted to their expected price development. Consequently, the price per kWh drops quickly to the respective mean reversion level until 2030 and remains relatively stable thereafter.

Appendix B. Analysis of extreme adoption shares

We conduct a separate analysis of extreme adoption shares by technology and across the relevant iterations (Figure B.4). First, we identify periods with a large spread of adoption. We investigate the iterations with the five highest or lowest rates for these periods. This analysis helps to understand which parameters can cause divergent technology adoption. Furthermore, it identifies useful adjustments to the average fleet renewal policy that can be applied if extreme (i.e., at the upper or lower end of the stochastic process spectrum) parameter developments are observed in practice.

The adoption probability for ICEVs is broadly spread in various periods. Especially conspicuous are the periods 2029 and 2044 immediately before the constraints. We see that a substantial deviation from the average adoption is often driven by extreme developments in vehicle purchase prices and fuel prices. In

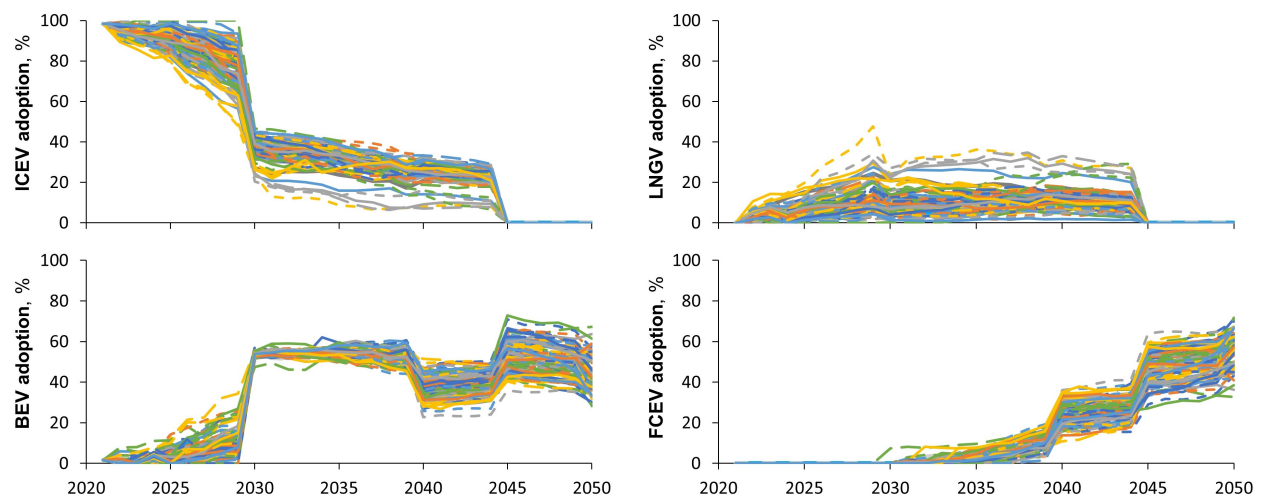


Figure B.4: ICEV (top left), LNGV (top right), BEV (bottom left), and FCEV (bottom right) adoption of all relevant iterations

2029, only the comparison of ICEV with LNGV costs is decisive. The other technologies play a subordinate role. ICEV adoption is high when LNGV vehicle or fuel prices are high or when ICEV prices or diesel prices are low. The opposite is true for low adoption in 2029. An nEHS price at the upper part of the stochastic process spectrum can also trigger lower adoption. As described, the nEHS price is used to monetarize the end-to-end carbon emissions in the target function. In 2044, low ICEV adoption can also be caused by a high nEHS price. Furthermore, extreme developments in the purchase and fuel prices can also make a difference. However, in 2044 the ICEV competes against all other technologies (and not only against the LNGV). If one of them experiences a low-cost development, it may weaken the adoption of the ICEV. The LNGV adoption probabilities are less spread out, but there is a hand full of iterations in which they reach 30 to 40%. Like before, they are primarily driven by the relative purchase and fuel price developments of ICEV and LNGV. Very high carbon prices also work in favor of the LNGV in direct comparison to the ICEV. The BEV adoption shows multiple interesting extreme points. An adoption of up to 34% can be achieved in 2029 when BEV purchase and electricity prices are low, but the purchase and fuel prices of the competing ICEVs and LNGVs are relatively high. FCEV cost parameters do not make a difference at this point. In 2044, the BEV adoption can take on a very low value if ICEVs and LNGVs perform well while the BEV purchase and electricity costs are high. Then, the emitting technologies absorb percentage points from the BEV technology, and the BEV cannot compete with the FCEV anymore. A critical observation can be made in 2050, where BEVs exclusively compete with FCEVs. The purchase prices are less important. As described before, we assume the mean reversion levels of both stochastic processes to be connected because of both technologies' similarities. The difference between them is not big enough to clearly influence the decision. Instead, the outcome is significantly driven by the nEHS price. Both technologies cause considerable carbon emissions in truck production. However, BEV production is evidently emitting more CO_{2e} than FCEV production, mainly caused by battery manufacturing (Booto et al., 2021; Rial & Pérez, 2021). We infer that the formulation of our target function benefits the FCEV in comparison with the BEV by accounting for the end-to-end emissions. On top of that, electricity and hydrogen prices can make a difference. If electricity prices are very low and hydrogen prices are very high, the BEV is adopted more and vice versa. Similar observations are made by looking at the extreme points of FCEV adoption. In 2044, the nEHS price and the relative price developments for all technologies are driving extreme adoption shares. In 2050, it is again the nEHS price and electricity and hydrogen prices.

Appendix C. Additional details related to the sensitivity analysis

Table C.1 summarizes the 11 tested sensitivities including a brief summary of their implementation. Complementary to that, Figure C.5 shows the respective modeled parameter developments. At this point, we describe the results of the less significant sensitivities that have been omitted in the main body of the publication.

Affected parameter	Number	Description	Implementation
Diesel price	S1	High level	- In 2022, current average market price of 2€ is assumed - Mean reversion level increased by gap in 2022 to base model average - New mean reversion level is 2.34€/l
	S2	Very high level	- As above, but mean reversion level increased by two times difference in 2022 - New mean reversion level is 2.87€/l
Hydrogen price	S3	Company-internal production	- Follows base model until 2024, as internal production not ready yet - Between 2025 and 2029 price between 4.5€/kg and 5.5€/kg is assumed - From 2030, price reaches internal target of 4.5€/kg with a certain variance
CO2e price (EU ETS & German nEHS)	S4	Lower level	- EU ETS mean reversion level is average from literature (94€/tCO2e) - nEHS mean reversion level decreased in same relation to 175€/tCO2e
	S5	Higher level	- EU ETS mean reversion level increased by difference from S4 to 166€/tCO2e - nEHS mean reversion level is decreased in the same relation to 278€/tCO2e
	S6	Faster maturity	- Mean reversion speed is set to 0.3 - As a result, mean reversion level is reached around 2030 (5 years earlier)
	S7	Slower maturity	- Mean reversion speed is set to 0.125 - As a result, mean reversion level is reached around 2040 (5 years later)
BEV purchase price	S8	Faster maturity	- Mean reversion speed is set to 0.5 - As a result, mean reversion level is reached around 2025 (5 years earlier)
	S9	Slower maturity	- Mean reversion speed is set to 0.175 - As a result, mean reversion level is reached around 2035 (5 years later)
Distance to be covered	S10	Smaller growth	- Total annual distance to be covered grows by one percent less (2% p.a.)
	S11	Larger growth	- Total annual distance to be covered grows by one percent more (4% p.a.)

Table C.1: Description of tested sensitivities

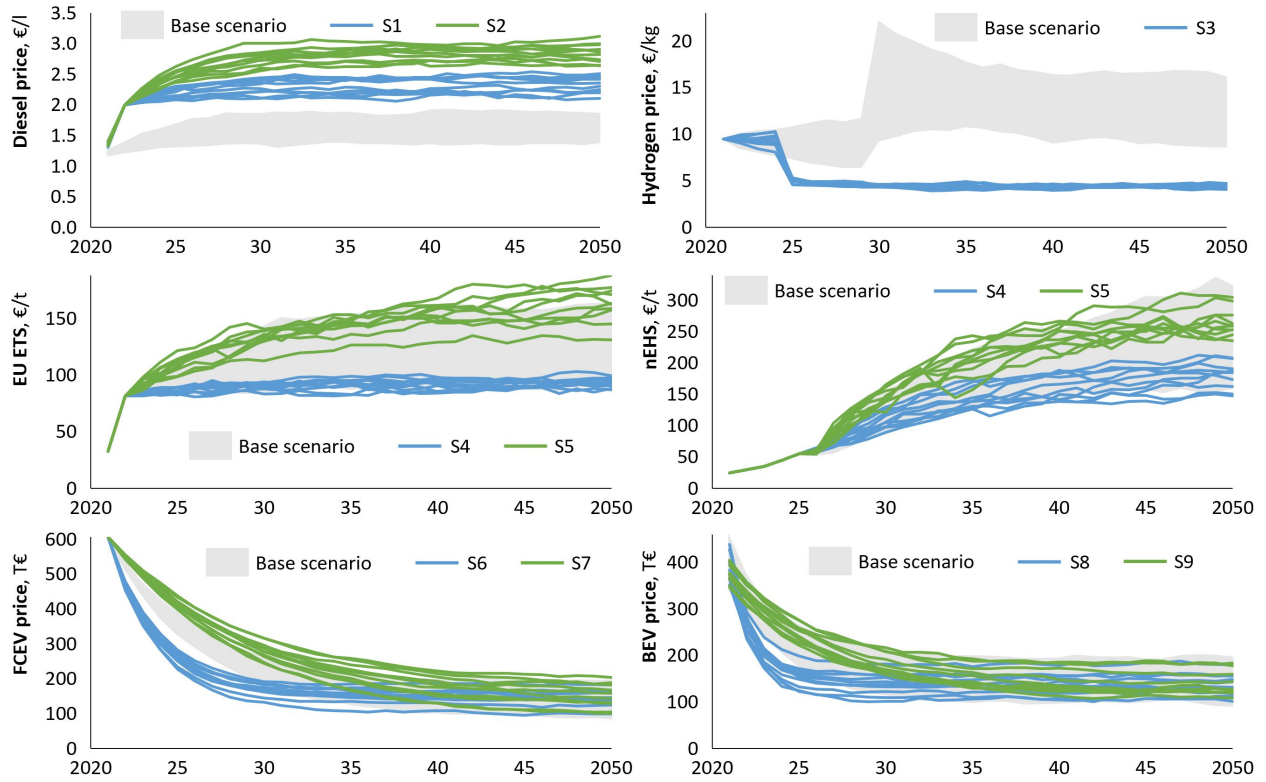


Figure C.5: Sample parameter developments in the sensitivity scenarios compared to the base scenario

The carbon price scenarios S4 (lower price level) and S5 (higher price level) only make a minimal impact on the adoption from 2040 onwards (below two (S4) and 4% (S5)). Interestingly, they influence only the BEV and the FCEV adoption, which is a counter-intuitive result at first sight. As a matter of fact, this again emphasizes that the algorithm prefers the lower cost position of the combustion engine technologies. The (even much higher) carbon costs are not big enough to significantly alter this strategy. On the contrary, our target function formulation even benefits ICEV and LNGV to some extent. Due to the small annual distances covered in the case study, fixed costs are highly important in the technology decision. Naturally, they are (currently) much higher for the alternative technologies. On top of that, we rightfully consider the end-to-end CO₂e emissions in the target function. ICEVs and LNGVs also have very low production emissions compared to the alternative technologies. Instead, they emit much more carbon equivalents during the use phase, where the alternative technologies are carbon-free (Booto et al., 2021; Rial & Pérez, 2021). We deduce that practitioners should always consider end-to-end emissions and keep their operational characteristics in mind when looking for technological alternatives to decrease emissions. A hasty switch to BEVs or FCEVs does not necessarily make sense from a holistic perspective, except the carbon footprint of BEV and FCEV production decreases significantly.

Equally, we investigate a faster and slower BEV purchase price maturity as part of scenarios S8 and S9. The effect of the two sensitivities is much less significant. We observe a slight effect on the adoption of

BEVs and ICEVs between 2023 and 2039. In the faster scenario, the BEV inherits a maximum of 3% from the ICEV, and in the slower scenario, the BEV loses not more than 4% to the ICEV. In the later periods, the faster maturity does not cause a significant change at all. The delayed maturity changes adoption after 2040, when some FCEVs are used instead of BEVs. However, the size of the difference fluctuates only between two to 4%. Evidently, the speed of BEV purchase price maturity does not make an enormous difference. We conclude that, in any case, the adoption of the BEV is essentially driven by the first emissions constraint and not by timely cost competitiveness with the ICEV. Furthermore, also due to the constraint, the FCEV cannot significantly benefit from a slower BEV price maturity. The FCEV maturity speed is still too slow to compete with the BEV in 2030. When the FCEV finally reaches a competitive price, the BEV has already become the dominant technology in the fleet, the BEV prices have sunken considerably, and as a result, the FCEV only receives a few additional percentage points.

Lastly, we examine the effect of alternative demand growth (i.e., growth of annual distance driven) assumptions on technology adoption. The chart displaying the results of scenarios S10 and S11 indicates that there is only a visible impact between 2030 and 2044 when the percentual constraint is effective. We conclude that the average policy remains unchanged. Again, the algorithm tries to deploy ICEVs and LNGVs as much as the emissions constraints allow. While the emissions constraint stays the same in the scenarios, the total distance driven and the fleet size do not. A higher demand growth requires a larger total fleet. Hence, the share of ICEVs and LNGVs in the respective periods shrinks. The opposite is true for smaller demand growth.

Appendix D. Overview of model parameters and sources

Category	Sub category	Parameter	Unit	Sources
Price parameters - Stochastic processes	Vehicle purchase prices	ICEV purchase price	€/truck	Lastauto Omnibus-Katalog (2018) Gnann et al. (2017) Giuliano et al. (2021) Kühnel et al. (2018) Mottschall et al. (2019) Kasten et al. (2016) Hacker et al. (2014)
		LNG purchase price	€/truck	Gnann et al. (2017) Kühnel et al. (2018) Mottschall et al. (2019) Kasten et al. (2016) Hacker et al. (2014)
		BEV purchase price	€/truck	Gnann et al. (2017) Giuliano et al. (2021) Kühnel et al. (2018) Unterlohner (2020)
		FCEV purchase price	€/truck	Gnann et al. (2017) Kühnel et al. (2018) Mottschall et al. (2019) Kasten et al. (2016) Hacker et al. (2014) Unterlohner (2020)
		Diesel price (provision)	ct/kWh	Kühnel et al. (2018) Mottschall et al. (2019)
	Fuel prices	LNG fuel price (provision)	ct/kWh	Kühnel et al. (2018) Mottschall et al. (2019)
		Electricity price (provision)	ct/kWh	Mottschall et al. (2019)
		Hydrogen price (provision)	ct/kWh	Kühnel et al. (2018)
				Mottschall et al. (2019)

Category	Sub category	Parameter	Unit	Source
	Carbon prices	EU ETS price	€/t	Prognos et al. (2020) Deutscher Bundestag (2020) Kemmler et al. (2020) Pyrka et al. (2020) Prognos et al. (2020) Prognos et al. (2021)
		nEHS price	€/t	Umweltbundesamt & DEHSt (2020) Prognos et al. (2020) Hennes et al. (2021)
	Battery purchase price	BEV battery	€/kWh	He et al. (2020) Mauler et al. (2021)
Price parameters - Deterministic processes	Infrastructure costs	Infrastructure costs for BEVs and FCEVs	€	Case study data
	Purchase fix costs	Purchase fix costs for all technologies	€	Case study data
	Maintenance costs	ICEV maintenance costs	€/km	Gnann et al. (2017)
		LNG maintenance costs	€/km	Gnann et al. (2017)
		BEV maintenance costs	€/km	Wietschel et al. (2019) Gnann et al. (2017) Wietschel et al. (2019)
		FCEV maintenance costs	€/km	Gnann et al. (2017)
	Tax costs	Tax costs for each technology	€/year	Bundesministerium der Justiz (2002)
	Toll costs	Toll costs for each technology	€/km	Toll Collect (2022)
	Driver costs	Driver cost across all technologies	€	Case study data
	Funding	Funding receivable	%	BMVI (2021)
		Individual limit per vehicle	€	BMVI (2021)
		Total limit per period	€	BMVI (2021)
Emissions factors	Scope 1	CH4 and N2O emissions of a heavy-duty ICEV CO2 emissions of a heavy-duty ICEV	g/km kg/l	Ntziachristos et al. (2009) Amon et al. (2019)
	Scope 2	CH4 and N2O emissions of a heavy-duty LNG CO2 emissions of a heavy-duty LNG	g/km kg/l	EPA US (2021) Amon et al. (2019)
		CO2e emissions from various energy sources	g/kWh	Huckestein (2021)
		CO2 emissions from various energy sources	kg/TJ	Icha et al. (2021)
	Scope 3	CH4 and N2O emissions from various energy sources	kg/TJ	Eggleson et al. (2006)
		CO2e emissions ICEV production, maintenance, end of life	kg/km	Rial & Pérez (2021) Booto et al. (2021)
		CO2e emissions LNG production, maintenance, end of life	kg/km	Rial & Pérez (2021)
		CO2e emissions BEV production, maintenance, end of life	kg/km	Booto et al. (2021)
		CO2e emissions FCEV production, maintenance, end of life	kg/km	Booto et al. (2021)
		CH4, CO2, N2O emissions of diesel & LNG fuel production	Gg/10 ³ m ³	Eggleson et al. (2006)
		CO2e emissions of energy source production for electricity	various	Eggleson et al. (2006) Amon et al. (2019)
		CO2e emissions of hydrogen production	kg/kg	Prognos et al. (2021) Kemmler et al. (2020) Valente et al. (2021)
	Other	100-year global warming potential of CO2, CH4, N2O	none	Eggleson et al. (2006)
Technology performance parameters	Fuel consumption	ICEV diesel fuel consumption rate	l/100km	Case study data Gnann et al. (2017)
		LNG fuel consumption rate	l/100km	Gnann et al. (2017)
		BEV electricity consumption rate	kWh/100km	Gnann et al. (2017)
		FCEV hydrogen consumption rate	kg/100km	Gnann et al. (2017)
	Tank/battery capacity	ICEV tank volume	l	Eurotransport (2013)
		LNG tank volume	kg	Knauber (2022)
		Electric battery capacity	kWh	Giuliano et al. (2021)
		FCEV tank volume	kg	Hyundai Motor Deutschland GmbH (2020)
	Refueling speed	ICEV refueling speed	l/minute	Gnann et al. (2017)
		LNG refueling speed	kg/minute	Eurotransport (2013)
		FCEV refueling speed	kg/minute	Knauber (2022)
	Residual value	Residual value ICEV, LNG	€	Hyundai Motor Deutschland GmbH (2020) ADAC e.V. (2022)
		Residual value BEV	€	Eurotransport (2013) ADAC e.V. (2022)
		Residual value FCEV	€	E-Force One AG (2018) Designwerk Products AG (2020) Hyundai Motor Deutschland GmbH (2020)
Operations parameters	Operating model	E.g., Avg. speed, distance driven, shift model (by cluster)	various	Case study data
	Use period	Expected use periods ICEV, LNG, BEV, FCEV	years	Case study data
		Expected distance until replacement electric battery	km	Kühnel et al. (2018)
		Expected distance until replacement fuel cell	km	Gnann et al. (2017)
	Purchase budget	Purchase budget by cluster	€	Case study data
Other	Other	Fuel value for diesel, LNG, hydrogen	kWh/l, kWh/kg	World Nuclear Association (2022)
		Energy mix for electricity generation in Germany	%	Icha et al. (2021)
		Margin of truck producers	%	Prognos et al. (2021) Hubik (2020)

Category	Sub category	Parameter	Unit	Source
		Margin of battery manufacturers	%	ecomento (2020)
		Margins for diesel and LNG production and distribution	ct/l	Statista (2021)
				Sürig (2014)
		Margin of electricity producers	ct/kWh	Götz et al. (2013)
		Margin of hydrogen producers	%	Horváth & Partners (2020)
		Added value tax	%	mehrwertsteuerrechner (2022)
		German energy tax	ct/l	Bundesministerium der Justiz (2006)
		German electricity tax	ct/kWh	Bundesministerium der Justiz (1999)
		EEG reallocation charge	ct/kWh	Hoffmann & Rottländer (2022)
		Inflation	%	finanz-tools (2022)

Table D.2: Overview of model parameters and sources

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